

## Formulas & Definitions

This page explains all calculations and metrics used in the Customer Analytics and Coupon Analytics dashboards.

### Key Metric Definitions

#### Period Metrics

##### **New Members (Period)**

Customers whose Registration Date falls within the selected period

##### **Returning Customers (Period)**

Count of customers with at least 1 return visit in the period

##### **Active (Period)**

Total unique customers (VIP IDs, excluding VIP ID = 0) who made purchases in the period

##### **Return Visit Rate (Period)**

$(\text{Returning Customers} / \text{Active Customers}) \times 100$

##### **Invoices (Customers)**

Total invoices from customers with valid VIP IDs (excludes VIP ID = 0)

##### **Amount (Customers)**

Sum of sales\_amount for transactions with valid VIP IDs (excludes VIP ID = 0)

##### **Total Invoices**

All invoices in period including VIP ID = 0

##### **Total Amount**

Sum of all sales\_amount in period including VIP ID = 0

#### All-Time Metrics

##### **Total Customers (All Time)**

Total count of records in Customer table

##### **Member Active (All Time)**

Customers with points > 0

##### **Member Inactive (All Time)**

Customers with points = 0

##### **Return Visit Rate (All Time)**

$(\text{Returning Customers} / \text{Total Customers in DB}) \times 100$

## 🔗 Customer Analytics Formulas

### 1. Return Visits Calculation ⚠️ CRITICAL FORMULA

The Return Visits metric counts how many invoices a customer had AFTER their first visit.

⚠️ **IMPORTANT:** This formula considers the **Registration Date** to determine if the first purchase is a "first visit" or a "return visit".

#### Complete Formula:

```
IF first_purchase_date == registration_date:
    return_visits = total_purchases - 1
    # First invoice on reg day = first visit (NOT a return)
    # All subsequent invoices = return visits

ELSE (first_purchase_date != registration_date OR no reg date):
    return_visits = total_purchases
    # Customer already visited to register
    # ALL purchases = return visits

is_returning = (return_visits > 0)
Note: We count INVOICES, not unique days
```

**Key Insight:** The formula is applied **CONSISTENTLY** at all levels (Global, Shop, Season), but uses **DIFFERENT purchase lists** for each context, resulting in **different "first\_purchase\_date"** values.

#### Example 1: Registration day purchase (Same Day)

- Registration Date: **2026-01-01**
- Purchases: **2026-01-01** (3 invoices on same day)
- first\_date == reg\_date ☒
- **Return Visits = 3 - 1 = 2**
- Is Returning = True (had 2 return visits on reg day)

**Example 2:** Registered before, buying later (Different Day)

- Registration Date: **2026-01-01**
- Purchases: **2026-02-15** (2 invoices)
- first\_date != reg\_date 
- **Return Visits = 2** (all purchases are returns)
- Is Returning = True

**Example 3:** Single purchase on registration day

- Registration Date: **2026-01-01**
- Purchases: **2026-01-01** (1 invoice)
- first\_date == reg\_date 
- **Return Visits = 1 - 1 = 0**
- Is Returning = False (no returns yet)

**Example 4:** No registration date

- Registration Date: **NULL**
- Purchases: **2026-03-10** (2 invoices)
- first\_date != reg\_date  (NULL comparison)
- **Return Visits = 2** (assume already visited to get VIP ID)
- Is Returning = True

 **Context-Dependent Calculations**

**CRITICAL:** The same customer can have DIFFERENT returning status in different contexts!

 **Why Totals Don't Match:** This is **CORRECT** behavior, not a bug!  
 Global returning invoices ≠ Sum of shop returning invoices ≠ Sum of season returning invoices

**Real Example - Customer A:**

**Registration:** 2026-01-01

**Purchases:**

- Invoice #1: Shop X, 2026-01-01 (reg day)
- Invoice #2: Shop Y, 2026-02-01

- Invoice #3: Shop C, 2026-03-01
- Invoice #4: Shop C, 2026-03-01

| CONTEXT                                 | PURCHASES  | FIRST DATE | REG DATE | FORMULA     | RETURN VISITS | IS RETURNING? | RET INVOICES     |
|-----------------------------------------|------------|------------|----------|-------------|---------------|---------------|------------------|
| <b>GLOBAL</b>                           | 4 invoices | 1/1        | 1/1      | $4 - 1 = 3$ | 3             | YES           | <b>4</b>         |
| Shop X                                  | 1 invoice  | 1/1        | 1/1      | $1 - 1 = 0$ | 0             | NO            | 0                |
| Shop Y                                  | 1 invoice  | 2/1        | 1/1      | $1 (!=)$    | 1             | YES           | 1                |
| Shop C                                  | 2 invoices | 3/1        | 1/1      | $2 (!=)$    | 2             | YES           | 2                |
| <b>TOTALS:</b>                          |            |            |          |             |               |               |                  |
| Global Returning Invoices:              |            |            |          |             |               |               | <b>4</b>         |
| Sum of Shop Returning Invoices (X+Y+C): |            |            |          |             |               |               | <b>0+1+2 = 3</b> |
| <b>Difference:</b>                      |            |            |          |             |               |               | <b>1 invoice</b> |

### Why Different?

- **Shop X:** First purchase at Shop X on reg day ( $1/1 == 1/1$ ) → NOT returning at Shop X
- **Shop Y:** First purchase at Shop Y after reg day ( $2/1 != 1/1$ ) → IS returning at Shop Y
- **Globally:** First purchase on reg day ( $1/1 == 1/1$ ), but has 4 total → IS returning globally

✓ **This is CORRECT!** Same customer, different "first\_purchase\_date" per context.

## 2. Return Visit Rate

Return Visit Rate (Period) % = (Returning Customers / Active Customers) × 100

- **Returning Customers:** Number of customers with return\_visits > 0
- **Active Customers:** Total unique customers who made purchases in the period

**Example:** In January 2026:

- 100 unique customers made purchases
- 75 of them have return\_visits > 0
- **Return Visit Rate =  $(75 / 100) \times 100 = 75\%$**

## 3. VIP ID = 0 (Buyer Without Info)

Purchases with **VIP ID = 0**, blank, or null are **EXCLUDED** from customer metrics but **INCLUDED** in total invoice/amount metrics.

**Invoices (Customers)**

Total invoices from customers with valid VIP IDs (excludes VIP ID = 0)

**Amount (Customers)**

Sum of sales\_amount from customers with valid VIP IDs (excludes VIP ID = 0)

**Total Invoices**

ALL invoices including VIP ID = 0

**Total Amount**

Sum of ALL sales\_amount including VIP ID = 0

**Example:**

- 100 customers made 500 invoices totaling 250M VND
- 50 invoices (25M VND) have VIP ID = 0

**Results:**

- Active (Period): 100 (excludes VIP 0)
- Invoices (Customers): 450
- Amount (Customers): 225M VND
- Total Invoices: 500 (includes VIP 0)
- Total Amount: 250M VND (includes VIP 0)

**4. Season (Year-Aware)**

Seasons are defined as:

- **M2-4** (Feb–Apr): February, March, April
- **M5-7** (May–Jul): May, June, July
- **M8-10** (Aug–Oct): August, September, October
- **M11-1** (Nov–Jan): November, December, January (spans 2 years)

Each season is labeled with the year(s) it covers:

**Examples of year-aware season labels:**

- **M2-4 2025**: Feb–Apr 2025

- **M11-1 2025-2026**: Nov 2025 + Dec 2025 + Jan 2026
- **M11-1 2024-2025**: Nov 2024 + Dec 2024 + Jan 2025

For date range Jan 2025 to May 2026, the "By Season" table shows all season windows with data:

- M11-1 2024-2025 (if Jan 2025 has data)
- M2-4 2025
- M5-7 2025
- M8-10 2025
- M11-1 2025-2026
- M2-4 2026
- M5-7 2026 (if May 2026 has data)

## 5. VIP Grade Order

Grades are always sorted in this order:

1. No Grade
2. Member
3. Silver
4. Gold
5. Diamond

Note: "Olden" and "Golden" are automatically normalized to "Gold"

## Coupon Analytics Formulas

### 1. Coupon Usage Percentages

By Shop Table shows two percentage calculations:

**% Used / Total (Period):**

= (Used Coupons at Shop / Total Coupons in Period - All Shops) × 100

**% Used / Total Used (All Shops):**

= (Used Coupons at Shop / Total USED Coupons - All Shops) × 100

**Example:** Period has 1000 coupons total across all shops, 400 are used.

- Shop A used 100 coupons
- **% Used / Total (Period) = (100 / 1000) × 100 = 10%**
- **% Used / Total Used (All Shops) = (100 / 400) × 100 = 25%**

## 2. Coupon ID Prefix Filter

The prefix filter is **case-insensitive** and matches the beginning of the Coupon ID.

**Example:** Filter = SEMIR50K

Matches:

- SEMIR50K001
- SEMIR50K002
- semir50k003 (case-insensitive)

Does NOT match:

- ABC-SEMIR50K (prefix must be at start)
- SEMIR40K001 (different prefix)

## 3. Customer Name & Phone in Coupon Detail

When a coupon is used, we lookup the customer's name and phone number using the VIP ID from the matched invoice.

- If no invoice match: VIP ID, Name, Phone are blank
- If invoice found but VIP ID = 0: Name = "-", Phone = "-"
- If invoice found with valid VIP ID: Name and Phone from Customer table

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